

Tyler Vander Mooren

ITC 413: NoSQL Databases

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Assignment 6

Option A Reconnection Analysis

```
1 MATCH (me:Person {name: "Tyler Vandermooren"})-[r:KNOWS]-  
  (contact:Person)  
2 WHERE r.last_contact IS NOT NULL  
3 WITH contact, r,  
4   date(r.last_contact) AS lastDate  
5 WITH contact, r,  
6   duration.inDays(lastDate, date()) AS d  
7 RETURN contact.name AS contact_name,  
8   r.last_contact AS last_contact_date,  
9   d.days AS days_since_contact,  
10  r.strength AS relationship_strength  
11 ORDER BY days_since_contact DESC;  
12
```

contact_name	last_contact_date	days_since_contact	relationship_strength
"Carol Davis"	"2023-01-01"	1143	1
"Alice Johnson"	"2023-01-01"	1143	5

Started streaming 2 records after 20 ms

Part 1 Query Description

This query was developed to identify individuals in my network with whom I have not interacted in more than six months. The query matches my Person node and traverses KNOWS relationships to connected contacts. Because the last_contact property is stored as a string, it is converted to a date format before calculating elapsed time. The duration.inDays() function is used to determine the total number of days since the last interaction. The output returns the contact's name, last contact date, total days since contact and relationship strength. Results are ordered in descending order of days since contact to prioritize the most overdue reconnections. This meets all assignment requirements by applying time-based filtering, including required fields and ordering results appropriately.

Part 2 Query Analysis

This query identifies dormant professional relationships by measuring elapsed time since last contact. An actionable outcome would be to reconnect with individuals whose interaction gap exceeds six months. Maintaining consistent communication supports long-term networking, which is especially important in cybersecurity and technology fields where referrals and collaboration are common.

Option B Network Connectors Analysis

```
1 MATCH (p:Person)
2 OPTIONAL MATCH (p)-[:KNOWS]-(direct:Person)
3 WITH p, COUNT(DISTINCT direct) AS direct_connection_count
4 OPTIONAL MATCH (p)-[:KNOWS]-(direct:Person)-[:KNOWS]-
  (extended:Person)
5 WHERE extended <> p
6 WITH p,
7     direct_connection_count,
8     COUNT(DISTINCT extended) AS total_network_reach
9 RETURN p.name AS person_name,
10     direct_connection_count,
11     total_network_reach
12 ORDER BY direct_connection_count DESC
13 LIMIT 3;
14
```

person_name	direct_connection_count	total_network_reach
"Tyler Vandermooren"	2	1
"Alice Johnson"	2	1
"Carol Davis"	1	1

Started streaming 3 records after 65 ms and completed after 67 ms.

Part 1 Query Description

To determine the most influential individuals within my network, I constructed a query that measures both direct and extended connections. The query counts direct connections using the KNOWS relationship and ensures uniqueness with COUNT(DISTINCT direct). It then calculates second-degree connections to determine total network reach by traversing from each

person to their direct contacts and then to those contacts' connections. The query returns each person's name, direct connection count, and total network reach. Results are ordered in descending order by direct connections and limited to the top three individuals. This satisfies the assignment requirements by demonstrating aggregation, relationship traversal, ordering, and result limitation.

Part 2 Query Analysis

This query highlights central individuals within my network who act as bridges between others. An actionable outcome would be to strengthen relationships with highly connected individuals to expand access to second-degree contacts. Understanding network structure allows for more strategic professional engagement rather than random outreach.

Option C Mentorship Opportunities

```
1 MATCH (me:Person {name: "Tyler Vandermooren"})-[:HAS_SKILL]->(mySkill:Skill)
2 WITH collect(mySkill.name) AS mySkills
3
4 MATCH (mentor:Person)-[:HAS_SKILL]->(skill:Skill)
5 WHERE r.proficiency = "Expert"
6 AND NOT skill.name IN mySkills
7 AND mentor.name <> "Tyler Vandermooren"
8
9 RETURN mentor.name AS potential_mentor,
10        skill.name AS skill_name,
11        r.proficiency AS proficiency_level;
12
```

potential_mentor	skill_name	proficiency_level
"Alice Johnson"	"JavaScript"	"Expert"

Part 1 Query Description

For option C, the query developed identifies potential mentors who possess expert-level skills that I do not have. The query first collects my current skills, then searches for other individuals connected to Skill nodes with a proficiency level of “Expert.” It excludes skills already in my profile and removes my own node from the results. The output includes the potential mentor’s name, skill name and proficiency level. This satisfies the assignment requirements by identifying skill gaps and potential mentorship opportunities aligned with professional growth.

Part 2 Query Analysis

The results provide clarity regarding specific areas where professional growth is possible. By identifying individuals with advanced expertise, the network can be leveraged for targeted mentorship rather than generalized advice. Using graph-based queries in this way supports deliberate skill development and aligns networking efforts with clearly defined professional objectives.